

EXECUTIVE BRIEF

From Precise Numbers to Defensible Decisions

Mathematical thinking and reproducible computation for leaders using statistics and artificial intelligence

Control	Value
Production Unit	PU-B01-C01 — Mathematical Thinking and Reproducible Computation
Series	SRAI Book 1 — Mathematical Foundations
Audience	Executives, policy leaders, programme directors and analytical managers
Evidence base	Controlled Chapter 1 and validated notebook M1_N01
Status	Human-edited draft v1.1 for owner review

My central point A number is useful to a decision-maker only when we can explain where it came from, reproduce the work and defend the assumptions behind it.

A note from the author

Over more than three decades working with official statistics, agricultural censuses, information systems and decision-support projects, I have seen the same problem in different forms. A result can be technically impressive and still be unsafe to use. Sometimes the weakness lies in the data. Sometimes it lies in an undocumented assumption, a changed file, an inappropriate objective or a model applied well beyond the setting in which it was tested.

This is why I begin SRAI with mathematical thinking rather than with a fashionable algorithm. Before asking what a model can calculate, we need to be clear about what we are representing, what we have left out and what decision the calculation is meant to support.

The practical problem

Institutions can now produce forecasts, risk scores and dashboards very quickly. AI can also generate a plausible explanation in seconds. Speed is welcome, but it creates a new management difficulty: the result may travel through the institution faster than anyone checks whether it is sound.

Consider an agricultural production forecast showing an 8 percent increase. Officials may already be discussing storage, imports or market measures when a second analyst discovers that the random seed was not recorded, an input file had changed, several districts were excluded without explanation and cultivated-area units were inconsistent. The 8 percent figure is still on the page, but the evidence beneath it has become uncertain.

The risk in plain terms A result that cannot be reconstructed should not be allowed to carry a major

budget, policy or public-communication decision.

What I would ask the analytical team

- A clear statement of the real decision and the population or system concerned.
- Documented variables, assumptions, objectives, constraints and excluded factors.
- Reproducible data, code, parameters, versions and random seeds.
- Separate evidence that the calculation is correct from evidence that the model is adequate.
- An explicit account of uncertainty, failure modes, distributional effects and implementation constraints.

The SRAI theory-to-decision chain

I use the following chain because it forces the analyst—and the decision-maker—to slow down at the right moments. Each step has a different purpose, and skipping one usually creates trouble later.

Stage	Executive control question
Reality	What real system, population or decision are we trying to understand?
Abstraction	What aspects matter, and what has been deliberately omitted?
Model	Which variables, relationships, assumptions, objectives and constraints represent the problem?
Algorithm	What finite procedure converts the model into a result?
Python / implementation	Can the procedure be executed consistently in a controlled environment?
Verification	Did we implement and solve the intended model correctly?
Interpretation	What does the output mean—and what does it not mean?
Decision	How should evidence, context, costs, risks and human responsibility be combined?

Core distinction Verification asks whether we solved the model correctly. Validation asks whether we constructed an adequate model of the relevant reality. A program can be verified and still be scientifically or operationally unsuitable.

Five elements every analytical proposal should expose

Element	What management should see
Variables	Inputs, states, decisions, outputs and latent constructs are not confused.
Assumptions	Conditions such as representativeness, stability, independence and measurement quality are stated and tested.
Objectives	The mathematical loss or target corresponds to the institution's real objective and values.

Element	What management should see
Constraints	Budgets, laws, fairness, capacity, time and data availability are represented.
Evidence	Observed data, external comparisons, sensitivity tests and domain expertise confront the model.

Why the objective matters

An algorithm will optimize the target we give it, not the public purpose we forgot to express. A model can reduce average error while performing badly in remote or vulnerable areas. It can report high accuracy while repeatedly missing a rare but costly event. For that reason, I believe senior management must understand and approve the real objective—not simply accept the metric selected by the technical team.

A practical decision assurance protocol

For an analysis that may influence money, services, regulation or public trust, I recommend a short assurance record. It need not become a large bureaucratic exercise. It should simply make the following controls visible before the recommendation is accepted.

Control	Required evidence	Warning sign
Problem definition	Decision, target population, time horizon and accountable owner	The team begins with a tool or model rather than a decision.
Data provenance	Source, version, coverage, exclusions, transformations and quality limitations	Analysts cannot reproduce the exact analytical dataset.
Model transparency	Variables, assumptions, objective, constraints and omitted factors	A polished output is offered without a model statement.
Computational reproducibility	Environment, dependencies, code, parameters, seed and execution record	Results depend on undocumented manual steps or hidden state.
Verification	Tests, invariants, boundary cases and independent calculations	The only evidence of correctness is that the code ran.
Validation	External evidence, sensitivity, subgroup, temporal and geographic testing	Performance is reported only on the development data.
Decision analysis	Costs, benefits, uncertainty, false alarms, missed events and distributional effects	One metric is treated as sufficient for the decision.

Questions I would put to an AI-generated answer

- Which statements are observations, which are model outputs and which are interpretations?
- What evidence would demonstrate that the answer is wrong or inadequate?
- Can another analyst reproduce the result from the recorded inputs?

- Which affected groups experience the largest error or risk?
- What decision would change if the uncertainty were larger than reported?
- Who remains accountable when the AI recommendation is accepted?

Governance principle AI can assist at every stage of the workflow. It cannot remove responsibility from any stage, and it cannot turn an undocumented calculation into knowledge.

What an institution can do without unnecessary bureaucracy

- Adopt the SRAI assurance protocol for high-impact forecasts, scoring systems and AI-supported decisions.
- Require reproducibility records as standard deliverables in analytical contracts and internal projects.
- Separate technical verification, domain validation and executive authorization.
- Maintain versioned evidence so a published number can be reconstructed after data or code changes.
- Train managers to question objectives, constraints and consequences—not only model accuracy.

Immediate implementation: a 30-day management agenda

Period	Action	Output
Days 1–5	Select one consequential analytical product and identify its accountable owner.	Defined decision, users and risk level
Days 6–10	Document data provenance, variables, assumptions, objective, constraints and exclusions.	One-page model statement
Days 11–15	Recreate the computational environment and rerun the analysis independently.	Reproducibility and verification record
Days 16–20	Test sensitivity, external evidence, subgroups, geography and time.	Validation findings and limitations
Days 21–25	Assess false alarms, missed events, costs, benefits and affected groups.	Decision consequence analysis
Days 26–30	Approve, revise, restrict or reject the analytical product and archive the evidence.	Named decision and controlled release record

What reproducibility does—and does not—guarantee

I insist on reproducibility because it makes an analytical claim inspectable. But reproducibility should not be oversold. Reproducing a result does not prove that the data are representative, that the causal assumptions are right, that the objective is fair or that the recommendation is sensible. It proves that we can reconstruct the analysis. Validation, professional knowledge and accountable judgment must do the rest.

What the companion notebook demonstrates

Notebook M1_N01 uses a deliberately simple electricity-demand example. It records the environment and random seed, checks the expected predictions and verifies the model's sensitivity numerically. It also names what the model leaves out: humidity, seasonality, prices, outages and nonlinear thresholds. That final step matters. Correct code is valuable, but it cannot rescue an inadequate representation of reality.

Decision checklist

- Can the result be reproduced from controlled inputs?
- Has mathematical and computational correctness been verified?
- Has adequacy for the intended reality and population been validated?
- Are assumptions, exclusions, uncertainty and limitations visible?
- Are objective and constraints aligned with institutional purpose and public responsibility?
- Is a named human authorized and accountable for the final decision?

My final message When I am shown a model result, I do not want to know only what number appeared on the screen. I want to understand the chain of reasoning and evidence that makes the number fit—or unfit—for the decision before us.

Source and control note

Developed from SRAI Book 1, Chapter 1, “Mathematical Thinking and Reproducible Computation,” and canonical notebook M1_N01. Version 1.1 incorporates author-background and voice-based editorial revision. It remains a controlled draft until Mbaye Kebe completes substantive review and approval.